

EC200U&EG91xU Series

QuecOpen(SDK) PWM Development Guide

LTE Standard Module Series

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About the Document

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1 Introduction

Quectel EC200U series and EG91xU family modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS. It is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see **document [1]**.

This document is applicable to QuecOpen® solution based on CSDK build environment. It introduces the development of PWM in application layer, including PWM API, development example and debugging of Quectel EC200U series and EG91xU family modules.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
-	EC200U Series
EG91xU	EG912U-GL
	EG915U Series

1.2. Multiplex Pins to PWM

EC200U series and EG91xU family modules support multiplexing four pins to PWM feature: PWM_LPG, PWM_PWT, PWM_PWL0 and PWM_PWL1. Contact Quectel Technical Support for details about the hardware design.

NOTE

Only the pin multiplexed to PWM_LPG can be set to the network indicator light.

2 PWM API

2.1. Header File

ql_pwm.h, the header file of PWM API, is in the `\components\ql-kernel\inc` directory. Unless otherwise specified, all the header files mentioned in this document are in this directory.

2.2. API Overview

Table 2: API Overview

Function	Description
<i>ql_pwm_open()</i>	Opens a PWM channel.
<i>ql_pwm_close()</i>	Closes a PWM channel.
<i>ql_pwm_lpg_enable()</i>	Enables PWM_LPG and configures pulse period and high level duration of LPG.
<i>ql_pwm_pwt_enable()</i>	Enables PWM_PWT and configures frequency dividing coefficient, frequency and duty cycle of PWT.
<i>ql_pwm_pwl_enable()</i>	Enables PWM_PWL and configures channel and duty cycle of PWL.
<i>ql_pwm_disable()</i>	Disables a PWM channel.

2.3. API Description

2.3.1. ql_pwm_open

This function opens a PWM channel.

- **Prototype**

```
ql_errcode_pwm ql_pwm_open(ql_PwmSel pwm_sel)
```

- **Parameter**

pwm_sel:

[In] PWM channel. See **Chapter 2.3.1.1** for details.

- **Return Value**

See **Chapter 2.3.1.2** for details.

2.3.1.1. ql_PwmSel

The enumeration of PWM channel types:

```
typedef enum
{
    PWM_PWT,
    PWM_LPG,
    PWM_PWL0,
    PWM_PWL1,
    PWM_MAX
}ql_PwmSel
```

- **Member**

Member	Description
<i>PWM_PWT</i>	PWM Channel. Output frequency and duty cycle are adjustable.
<i>PWM_LPG</i>	LPG Channel. Output frequency and duty cycle are fixed.
<i>PWM_PWL0</i>	PWL0 Channel. Output frequency is not fixed, duty cycle is adjustable.
<i>PWM_PWL1</i>	PWL1 Channel. Output frequency is not fixed, duty cycle is adjustable.
<i>PWM_MAX</i>	An invalid channel.

2.3.1.2. ql_errcode_pwm

PWM result codes indicate whether the function is executed successfully. An error cause is returned if the execution failed. The enumeration of result codes of PWM:

```
typedef enum
{
    QL_PWM_SUCCESS = QL_SUCCESS,

    QL_PWM_EXECUTE_ERR           = 1|QL_PWM_ERRCODE_BASE,
    QL_PWM_INVALID_PARAM_ERR,
    QL_PWM_FUNC_SET_ERR,
    QL_PWM_ACQUIRE_ERR,
    QL_PWM_START_ERR,
    QL_PWM_STOP_ERR,
    QL_PWM_REPEAT_OPEN_ERR,
    QL_PWM_REPEAT_CLOSE_ERR
}ql_errcode_pwm
```

● Member

Member	Description
QL_PWM_SUCCESS	Successful execution.
QL_PWM_EXECUTE_ERR	Failed execution.
QL_PWM_INVALID_PARAM_ERR	Parameter error.
QL_PWM_FUNC_SET_ERR	Failed to configure PWM.
QL_PWM_ACQUIRE_ERR	Failed to get PWM information.
QL_PWM_START_ERR	Failed to enable PWM.
QL_PWM_STOP_ERR	Failed to stop PWM.
QL_PWM_REPEAT_OPEN_ERR	Repeating opening a PWM channel error.
QL_PWM_REPEAT_CLOSE_ERR	Repeating closing a PWM channel error.

2.3.2. ql_pwm_close

This function closes a PWM channel.

- **Prototype**

```
ql_errcode_pwm ql_pwm_close(ql_PwmSel pwm_sel)
```

- **Parameter**

pwm_sel:

[In] PWM channel. See **Chapter 2.3.1.1** for details.

- **Return Value**

See **Chapter 2.3.1.2** for details.

2.3.3. ql_pwm_lpg_enable

This function enables PWM_LPG and configures pulse period and high level duration of LPG.

- **Prototype**

```
ql_errcode_pwm ql_pwm_lpg_enable(ql_LpgPer lpg_period, ql_LpgOnTime highlvl_time)
```

- **Parameter**

lpg_period:

[In] LPG pulse period. See **Chapter 2.3.3.1** for details.

highlvl_time:

[In] High level duration of LPG in pulse period. See **Chapter 2.3.3.2** for details.

- **Return Value**

See **Chapter 2.3.1.2** for details.

2.3.3.1. ql_LpgPer

The enumeration of LPG pulse periods:

```
typedef enum
{
    LPG_PER_125MS,
```

```

    LPG_PER_250MS,
    LPG_PER_500MS,
    LPG_PER_1000MS,
    LPG_PER_1500MS,
    LPG_PER_2000MS,
    LPG_PER_2500MS,
    LPG_PER_3000MS
}ql_LpgPer
    
```

● Member

Member	Description
<i>LPG_PER_125MS</i>	LPG pulse period: 125 ms.
<i>LPG_PER_250MS</i>	LPG pulse period: 250 ms.
<i>LPG_PER_500MS</i>	LPG pulse period: 500 ms.
<i>LPG_PER_1000MS</i>	LPG pulse period: 1000 ms.
<i>LPG_PER_1500MS</i>	LPG pulse period: 1500 ms.
<i>LPG_PER_2000MS</i>	LPG pulse period: 2000 ms.
<i>LPG_PER_2500MS</i>	LPG pulse period: 2500 ms.
<i>LPG_PER_3000MS</i>	LPG pulse period: 3000 ms.

2.3.3.2. ql_LpgOntime

The enumeration of high level durations of LPG in pulse period:

```

typedef enum
{
    LPG_ONTIME_UNDEFINE,
    LPG_ONTIME_15_6MS,
    LPG_ONTIME_31_2MS,
    LPG_ONTIME_46_8MS,
    LPG_ONTIME_62MS,
    LPG_ONTIME_78MS,
    LPG_ONTIME_94MS,
    LPG_ONTIME_110MS,
    LPG_ONTIME_125MS,
    LPG_ONTIME_140MS,
    
```

```
LPG_ONTIME_156MS,
LPG_ONTIME_172MS,
LPG_ONTIME_188MS,
LPG_ONTIME_200MS,
LPG_ONTIME_218MS,
LPG_ONTIME_234MS
}ql_LpgOtime
```

● Member

Member	Description
<i>LPG_ONTIME_UNDEFINE</i>	No high level.
<i>LPG_ONTIME_15_6MS</i>	High level duration: 15.6 ms.
<i>LPG_ONTIME_31_2MS</i>	High level duration: 31.2 ms.
<i>LPG_ONTIME_46_8MS</i>	High level duration: 46.8 ms.
<i>LPG_ONTIME_62MS</i>	High level duration: 62 ms.
<i>LPG_ONTIME_78MS</i>	High level duration: 78 ms.
<i>LPG_ONTIME_94MS</i>	High level duration: 94 ms.
<i>LPG_ONTIME_110MS</i>	High level duration: 110 ms.
<i>LPG_ONTIME_125MS</i>	High level duration: 125 ms.
<i>LPG_ONTIME_140MS</i>	High level duration: 140 ms.
<i>LPG_ONTIME_156MS</i>	High level duration: 156 ms.
<i>LPG_ONTIME_172MS</i>	High level duration: 172 ms.
<i>LPG_ONTIME_188MS</i>	High level duration: 188 ms.
<i>LPG_ONTIME_200MS</i>	High level duration: 200 ms.
<i>LPG_ONTIME_218MS</i>	High level duration: 218 ms.
<i>LPG_ONTIME_234MS</i>	High level duration: 234 ms.

2.3.4. ql_pwm_pwt_enable

This function enables PWM_PWT and configures frequency dividing coefficient, frequency and duty cycle of PWT.

- **Prototype**

```
ql_errcode_pwm ql_pwm_pwt_enable(uint8_t pwm_divider, uint32_t pwt_period, uint32_t pwt_duty)
```

- **Parameter**

pwm_divider:

[In] Frequency dividing coefficient. Range: 0–255.

pwt_period:

[In] Frequency. Range: 0–2047.

Output frequency = $APB / ((pwt_period \times 8 + 1) \times (pwm_divider + 1))$, APB = 200 MHz.

pwt_duty:

[In] Duty cycle. Range: 0–1023.

- **Return Value**

See **Chapter 2.3.1.2** for details.

2.3.5. ql_pwm_pwl_enable

This function enables PWM_PWL and configures channel and duty cycle of PWL.

- **Prototype**

```
ql_errcode_pwm ql_pwm_pwl_enable(ql_PwmSel pwm_sel, uint32_t pwl_duty)
```

- **Parameter**

pwm_sel:

[In] PWM channel. See **Chapter 2.3.1.1** for details.

pwl_duty:

[In] Duty cycle. Range: 0–255.

- **Return Value**

See **Chapter 2.3.1.2** for details.

2.3.6. ql_pwm_disable

This function disables a PWM channel.

- **Prototype**

```
ql_errcode_pwm ql_pwm_disable(ql_PwmSel pwm_sel)
```

- **Parameter**

pwm_sel:

[In] PWM channel. See **Chapter 2.3.1.1** for details.

- **Return Value**

See **Chapter 2.3.1.2** for details.

3 Development Example and Debugging

This chapter introduces how to use the above API for PWM development and simple debugging in application layer.

3.1. Development Example in Application Layer

3.1.1. Example Description

led_cfg_demo.c, the example file, is provided in the `\components\ql-application\peripheral` directory of Quectel EC200U series and EG91xU family QuecOpen SDK. The example file contains configuration and usage of PWM, including opening, closing, disabling and enabling a PWM channel. The entry function is *ql_ledcfg_app_init()* as shown below.

The example sets the flashing mode of the network indicator light according to the network state of the current (U)SIM card.

```
void ql_ledcfg_app_init(void)
{
    ...QIOSStatus err = QL_OSI_SUCCESS;
    ...ql_task_t ledcfg_task = NULL;

    ...err = ql_rtos_task_create(&ledcfg_task, 1024, APP_PRIORITY_NORMAL, "ql_ledcfgdemo", ql_ledcfg_demo_thread, NULL, 1);
    ...if (err != QL_OSI_SUCCESS)
    ...{
    ...    QL_LEDCFGDEMO_LOG("led.config.demo.task.created.failed");
    ...}

    ...err = ql_event_register(ledcfg_task, QUEC_LEDCFG_EVENT_IND);
    ...if (err != QL_OSI_SUCCESS)
    ...{
    ...    QL_LEDCFGDEMO_LOG("led.config.event.register.failed");
    ...}
}
```

Figure 1: Flashing Mode of the Network Indicator Light

3.1.2. Example Auto-start

The above example is enabled by default in `ql_init_demo_thread()` as shown below.

```
static void ql_init_demo_thread(void *param)
{
    ... QL_INIT_LOG("init_demo_thread_enter, param:0x%x", param);
    #if 0
    ... ql_gpio_app_init();
    ... ql_gpioint_app_init();
    #endif
    ... ql_ledcfg_app_init();
    #ifdef QL_APP_FEATURE_AUDIO
    ... ql_audio_app_init();
    #endif
    #ifdef QL_APP_FEATURE_LCD
    ... ql_lcd_app_init();
    #endif
    ... ql_nw_app_init();
    ... //ql_datacall_app_init();
    ... ql_osi_demo_init();
}
```

Figure 2: Example Auto-start

3.2. PWM Debugging

3.2.1. Preparation

PWM of EC200U series and EG91xU family modules can be debugged on Quectel LTE OPEN EVB. You can view information of the example after capturing the AP logs with cooltools. For details on log capture, see **document [2]**.

NOTE

Insert a (U)SIM card before you debug the example.

3.2.2. Debugging

`ql_ledcfg_app_init()` automatically starts after the module is booted. Through log information, you can see the flashing frequency of network indicator light and the on-off status of 4G indicator light. At the same time, you can see the actual on-off status through the indicator light of NET_MODE and NET_STATUS, and you can also use an oscilloscope to measure the corresponding pin signals to see if the settings are correct.

The log debugging information is shown below:

```
[ql_LEDCFGDEMO][ql_ledcfg_demo_thread, 118] led config fast twinkle [7]
[ql_LEDCFGDEMO][ql_ledcfg_demo_thread, 139] led config network is 4G
[ql_LEDCFGDEMO][ql_ledcfg_demo_thread, 118] led config fast twinkle [9]
[ql_LEDCFGDEMO][ql_ledcfg_demo_thread, 139] led config network is 4G
[ql_LEDCFGDEMO][ql_ledcfg_demo_thread, 123] led config speed twinkle [11]
[ql_LEDCFGDEMO][ql_ledcfg_demo_thread, 139] led config network is 4G
[ql_LEDCFGDEMO][ql_ledcfg_demo_thread, 118] led config fast twinkle [10]
[ql_LEDCFGDEMO][ql_ledcfg_demo_thread, 139] led config network is 4G
```

Figure 3: Log Debugging Information

NOTE

Because the pins in this example may be configured in other examples, make sure that only this example runs when you debug PWM.

4 Appendix References

Table 3: Related Documents

Document Name
[1] Quectel_EC200U&EG91xU_Series_QuecOpen(SDK)_Quick_Start_Guide
[2] Quectel_EC200U&EG91xU_Series_QuecOpen(SDK)_Log_Capture_Guide

Table 4: Terms and Abbreviations

Abbreviation	Description
(U)SIM	(Universal) Subscriber Identity Module
AP	Access Point
APB	Advanced Peripheral Bus
API	Application Programming Interface
EVB	Evaluation Board
GPIO	General-Purpose Input/Output
IoT	Internet of Things
LED	Light Emitting Diode
LTE	Long-Term Evolution
PC	Personal Computer
PWM	Pulse Width Modulation
RTOS	Real-Time Operating System
SDK	Software Development Kit

UART	Universal Asynchronous Receiver/Transmitter
USB	Universal Serial Bus
